Detecting AI-Generated Images of People

Andy Yan David Li
University of Maryland, College Park
7901 Regents Dr
College Park, MD 20742
ayan1@terpmail.umd.edu dli12312@terpmail.umd.edu

Abstract

As a result of innovation in the field of generative AI, the public now has widespread access to AI tools that can easily generate images and art, raising concerns about authenticity and misuse. These include ethical concerns on the use of AI in media, art, entertainment, and more. This paper presents the design and implementation of a machine learning model which can detect AI-generated images, making use of backpropagation and stochastic gradient descent, tested and trained on AI-generated images of human faces created by StyleGAN. In addition, we propose to add a Laplacian residual layer to include high-frequency channels, extracting high-frequency artifacts. This addition is shown to drastically improve the performance of the detection model, proving high-frequency artifact detection a promising path in improving AI detection models.

1. Introduction

Generative AI has become more and more prevalent in society, having a variety of civilian applications. One of these applications is the use of AI models to generate imagery, including fake art and artificial photographs. AI generated imagery is typically created using neural networks, using natural language processing to parse the user's prompt, then using a Generative Adversarial Network (GAN) or a Diffusion model to then generate the image. Training the models on many hundreds of millions of images enables detailed, accurate image generation that is often difficult to discern from real photographs or art.

Using AI to generate imagery is a source of concern among the populace, since AI has gotten advanced enough to be able to fool many people with its art, creating art shockingly detailed in a matter of seconds. While AI generated imagery can be beneficial in many ways, it also contributes to the spread of misinformation, infringes on the work of digital artists, and even casts doubt on the validity of photographic and video evidence altogether. This calls for an efficient and accurate method to distinguish between AI generated images and real images. We propose a computer vision oriented machine learning solution to this problem.

In this paper, we present a design and implementation of a Convolutional Neural Network (CNN) using stochastic gradient descent with momentum. Additionally, we make use of data augmentation to handle the data. This model enables us to classify imagery as AI generated or not. We focus on identifying AI-generated images of human faces, created by StyleGAN, as the training and test data. We hope to greatly improve upon the human ability to identify AI imagery and provide a straightforward model to both precisely and quickly distinguish between real and fake images, as well as establish high frequency artifacts extracted by Laplacian residual layers as a strong indicator.

The remainder of this paper is structured as follows: an exploration of necessary background information as well as relevant prior research done in this specific field, while also discussing our motivation for this project (section 2), then we present the design of the model as well as relevant implementation details (section 3), discuss and visualize the results of our classification model (section 4), move on with the conclusion (section 5), then finally detail the contributions of the authors to this project (section 6).

2. Background

The use of generative AI to create imagery has seen an exponential growth in prevalence in recent years. It has seen widespread civilian use, in even commonly used apps like Photoshop to enhance or correct images. This is the result of a history of innovation and technological development, making it a powerful yet double edged tool.

Early AI-generated images were primarily done using Generative Adversarial Networks. While these worked, they often included obvious imperfections, such as misgenerated features or incorrect textures, making it easy even for a human to detect. However, this has been greatly improved upon by the development of modern models such as StyleGAN and stable diffusion, which produce images that are extremely convincing. Human observers often struggle to distinguish these images from real ones, necessitating a tool that can reliably identify AI-generated images. Many researchers have leveraged computer vision to differentiate between real and fake images, providing a precedent to our research.

2.1. Previous research

One of the most important studies was done by Wang et al. in his 2020 experiment, where he trained a GAN model to classify fake images and achieved a high accuracy rate. His work found that training on ProGAN and StyleGAN2 generated images yielded the highest accuracy, revealing the presence of so-called "artifacts" on AI-generated images which are left by CNN models on their images. They additionally found that data augmentation allowed for more generalization, but drastically changing the subject of the images reduced performance significantly.

Another study by Chai et al. focused on how to make detectors generalize. By experimenting with altering various image properties, they found that detectors tend to focus on texture inconsistencies as opposed to evaluating the actual content of the image. This was further proven by an experiment in which shuffling image patches, destroying the overall structure but preserving local data, barely hurt the performance of the detector at all.

As more sophisticated diffusion models began to emerge, concerns were raised about the ability of previous detectors to perform on diffusion model generated images. Corvi et al. put this to the test, experimenting thoroughly with various detectors, attempting to test the ability of these detectors to generalize to unseen architectures. They found that detectors perform very poorly on images generated by new diffusion models, and lack the ability to generalize. They did however discover that diffusion models also possess distinctive artifacts, but due to their variety and subtlety, they are more difficult to detect.

Ohja et al. developed a baseline solution by creating a feature space not specifically catered towards real vs fake classification. By using nearest neighbor or linear probing on this space, detectors find far more success in generalizing. Additionally, they found that a common fingerprint does exist between the previous GAN models' images and the diffusion models, leaving room for future researchers to further optimize their models.

2.2. Motivation

AI generated content in general has been the subject of controversy since the beginning, with most people acknowledging the pressing ethical concern of AI images used for malicious purposes or being used to replace art or media done by real people. AI-generated imagery is already negatively affecting people's trust in news and media.

Additionally, as shown from previous research, AI image generation is constantly shifting and developing evasion techniques to avoid detection. From adding spectral regularizers to more subtle artifacts in diffusion models, AI detectors must be constantly created and updated to stay relevant to the newest AI generated images available. This motivates us to create an up to date image detector.

On top of this, from performing research on previous detectors, we found the existence of artifacts an avenue worth exploring. Therefore, we proposed to use a Laplacian residual layer to introduce high frequency channels, in hopes that these fingerprints or artifacts could potentially be related to the high frequency components. This is a new avenue of exploration that we believe could introduce a new path to improving AI image detection.

3. Design

3.1. Traditional architecture

We present a convolutional neural network (CNN) trained from scratch for binary image classification. Our pipeline employs data augmentation (rescaling, rotation, translation, zoom and horizontal flipping), he-normal weight initialization, batch normalization, dropout regularization, and stochastic gradient descent with momentum. Over 50 epochs, we monitor training and validation accuracy on 128×128 RGB images.

3.2. Data Preprocessing

To obtain the images, we generated 100 fake faces using StyleGAN's github repository. For the real images, we extracted 100 real faces from the public dataset at CelebA.

Now it's time to process the data. Each original pixel can be represented by $x \in [0, 255]$. We first apply a rescaling effect

$$x' = \frac{x}{255} \quad \big(x' \in [0,1]\big).$$

We then augment the data by applying a series of random rotations, translations, flips, and zoom. This is done to expose the model to more "distinct" examples without collecting new data, which helps reduce overfitting since our original dataset is limited.

We split 85% of the samples for training, and 15% for validation. This split aims to maximize learning while still providing a good check on overfitting.

3.3. Convolutional Blocks

Our CNN uses three convolutional blocks followed by a fully connected head. All convolutions use He-normal initialization

$$W_{ij} \sim \mathcal{N}\Big(0, \frac{2}{\text{fan\_in}}\Big).$$

where fan_in is the number of input channels · kernel area.

Each block applies Conv2D (convolution layer) to learn filters that respond to small spatial patterns. A single filter is applied across the entire image, greatly reducing the number of parameters compared to a fully connected layer. The operation is

$$(Z_k)_{i,j} = \sum_{c=1}^{C} \sum_{u=-m}^{m} \sum_{v=-m}^{m} W_{k,c,u,v}\, X_{c,i+u,j+v} \;+\; b_k$$

where W_k is the kth kernel, C is the number of input channels, and the kernel size is (2m + 1) · (2m + 1).

We then apply batch normalization, which stabilizes the learning process by providing a slight noise effect that can reduce overfitting.

The operation per channel is

$$\mu = \tfrac{1}{m} \sum_i x_i, \quad \sigma^2 = \tfrac{1}{m} \sum_i (x_i - \mu)^2$$

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \varepsilon}}$$

$$y_i = \gamma\, \hat{x}_i + \beta$$

where $\gamma,\ \beta$ are learned.

ReLU activation is used to allow the network to approximate non-linear decision boundaries.

Max-pooling was used to keep only the strongest activation in each window.

$$y_{i,j} = \max_{0 \le u,v < p}\ x_{pi+u,\, pj+v}.$$

Finally, we implement a dropout with rate p which prevents co-adaptation of neurons by randomly "dropping" units during training. This is a simple way to prevent the neural network from overfitting. The operation is given by

$$M_i \sim \text{Bernoulli}(1-p), \quad y_i = \frac{x_i\, M_i}{1-p}.$$

3.4. Classification

For the classification process, we first take the 3D feature-map tensors from the convolution stack which has shape (H, W, C) and flatten it into a single vector

$$x \in \mathbb{R}^n, \quad n = H \times W \times C.$$

We then process this vector by applying batch normalization, ReLU activation, and dropout as mentioned previously. Finally, we produce a single probability using the sigmoid function. This is given by

$$\hat{y} = \sigma(s) = \frac{1}{1+e^{-s}} \ \in (0,1),$$

which we interpret as the probability of the "positive" class.

3.5. Training Procedure

We use the binary cross entropy loss function, which measures how far the model's predicted probability p is from the true label.

$$\mathcal{L}(y,p) = -\big[y \log p \;+\; (1-y)\log(1-p)\big].$$

If y = 1, it penalizes low p values, while if y = 0, it penalizes high p values.

At each training step we update the weights using both stochastic gradient descent and a "velocity" that carries information from past updates

$$v_t = \mu\, v_{t-1} \;-\; \eta\, \nabla_w \mathcal{L},$$

where $\mu = 0.9$ is the momentum coefficient and $\eta = 0.01$ is the learning rate. The weights are updated using the formula

$$w_t = w_{t-1} + v_t.$$

To test for accuracy, our CNN model's prediction rule is such that

$$\hat{y} = [\, p > 0.5\,]$$

The accuracy over n samples is given by

$$\text{Acc} = \tfrac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big(\hat{y}_i = y_i\big),$$

We run 50 epochs through the training set. We use 64 images for training and 32 images for validation. Training and validation accuracy were logged per epoch.

3.6. Detector with Laplacian layer

The detection architecture remains largely similar to the traditional detector described above, but is adapted to support a Laplacian residual layer. Each input sample is a $256 \times 256 \times 4$ tensor obtained by concatenating the three RGB channels with a Laplacian of Gaussian residual that highlights edges and other high-frequency structures. The Laplacian is computed with a 3×3 kernel ($\sigma = 1$ px), and then normalized, ensuring that gradient magnitudes in the Laplacian branch remain in line with those of colour features in the other layers. We pass the Laplacian residual explicitly, encouraging the detector to focus on where the visual artifacts are likely to appear as opposed to having to pick up on the clues from scratch.

All the data preprocessing procedures, overall convolutional structure, and parameters remain constant, allowing us to test the effectiveness of the detector with the addition of the Laplacian layer.

4. Results and findings

4.1. Traditional architecture results

```
Epoch 50/50
3/3 ━━━━━━━━━━━━━━━━━━━━ 19s 5s/step - accuracy: 0.9523
loss: 0.1262 - val_accuracy: 0.5333 - val_loss: 0.8835
```

Fig 1. Results after 50 epochs (traditional architecture)

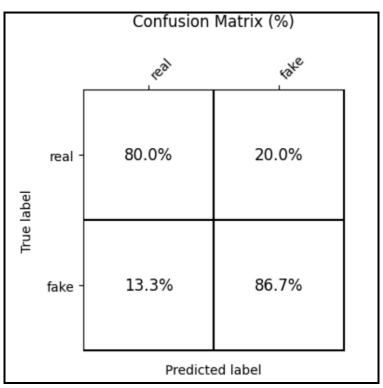

Fig 2. Confusion matrix (traditional architecture)

In Fig 2 above, 80 % are recognised as real, while 20 % are falsely identified as fake. In practical terms, one in five genuine images results in a false negative, which is not the most ideal result. For fake faces, 86.7 % are caught, and 13.3 % slip through as real.

From this, we observe that the model appears to be a little more willing to call an image fake than to call it real. This suggests that the decision threshold is adjusted to favor the fake class slightly.

```
Classification report:
              precision    recall  f1-score

        real       0.86      0.80      0.83
        fake       0.81      0.87      0.84

    accuracy                           0.83
   macro avg       0.83      0.83      0.83
weighted avg       0.83      0.83      0.83
```

Fig 3. Classification report (traditional architecture)

In Fig 3, the averages all converge at .83 for precision, recall, and f1 score. It is worth noting several per-class details, namely that precision is higher for real (0.86) than for fake (0.81). Whenever the model determines an image to be real, it is usually correct; however, its “fake” identifications are less likely to be correct. On the flip side, recall inverts that relationship: the detector recovers more fakes (0.87) than reals (0.80). This again points to a small bias toward predicting the fake class.

4.2. Addition of Laplacian layer

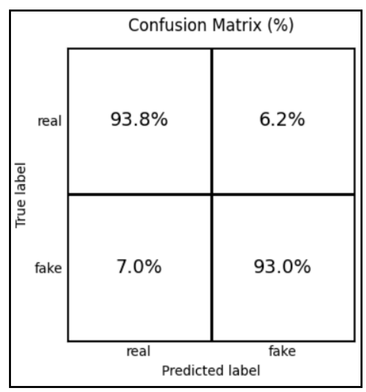

Fig 3. Confusion matrix (Laplacian layer)

The addition of the Laplacian layer results in a clear performance increase compared to the traditional architecture. The baseline incorrectly identified one in every five genuine images as fake, with the Laplacian channel that misidentification falls to 6.2%. The detector was already reasonably good at detecting fake faces, at 86.7%, but the Laplacian layer elevates it to 93%. Put simply, both types of errors have been reduced to less than 7%, a significant improvement.

Classification Report:

	precision	recall	f1-score
real	0.9510	0.9379	0.9444
fake	0.9118	0.9300	0.9208
accuracy			0.9347
macro avg	0.9314	0.9340	0.9326
weighted avg	0.9350	0.9347	0.9348

Fig 4. Classification report (Laplacian layer)

The classification report tells a similar story. For real images the F1‑score increases from 0.83 to 0.94 due to the large increase in recall, while for fake images it moves from 0.84 to 0.92 with both precision and recall increasing. The macro‑averaged F1 improves by 10% - from 0.83 to 0.93, indicating a large overall improvement in detection accuracy.

4.3. Discussion

The results presented in Fig 1–4 highlight the effectiveness of modelling high‑frequency components in the detection architecture. When the network is limited to conventional RGB information (Fig 1–2), it achieves an overall accuracy of 0.83, but the diagonal of the confusion matrix is lower than ideal: 20 % of genuine photographs are misclassified as synthetic and 13.3 % of fake images are undetected. These errors suggest that the baseline relies solely on color statistics and low‑frequency texture. These cues, while informative and worth considering, are prone to confounding by noise such as blur, JPEG quantisation or sensor grain.

Introducing the Laplacian residual channel (Fig 3–4) reduces error significantly. The macro‑averaged F1‑score climbs by a full tenth of a point, from 0.83 to 0.93, confirming that the increase in performance is symmetric across both classes. Because the Laplacian isolates edges and other high‑frequency artifacts present through the upsampling and aliasing techniques used by generative AI, the convolutional filters are presented with a map of where generative discrepancies are likely to occur. This allows the model to learn a cleaner decision boundary without having to rediscover those cues from raw RGB.

Through this study, we can conclusively determine that many distinctive clues lie in the high-frequency components of AI-generated images. This is crucial to take into consideration when building a model to classify imagery generated by AI. A Laplacian residual layer is a simple yet effective method to extract these clues and build them into the model, significantly increasing performance. The result supports the idea that priors and learned features are not mutually exclusive. When combined, they yield a better performance compared to using either approach separately.

5. Conclusion

In this project, we proposed a binary classifier to distinguish fake (AI-generated) photos of people from real photographs. Our architecture builds a convolutional neural network from scratch and incorporates a Laplacian filter layer up front to isolate high-frequency components. Adding the Laplacian layer led to a noticeable boost in recall and F1 score, showcasing how leveraging high-frequency artifacts is a powerful component to conventional CNN feature learning.

Many improvements could be made to this project: we used a small dataset with only 100 real and fake images, and they all came from a single source. Expanding into cross-domain image sets is another option, which would contribute to generalization. Through systematic validation, tuning, and architectural refinements, the CNN's resilience and performance can be substantially improved.

By demonstrating that a computationally inexpensive Laplacian channel can unlock a double‑digit improvement in accuracy, we hope to offer proof that high-frequency analysis of images shows promise in greatly increasing performance for detecting AI generated images. This implementation is a demonstration that while larger datasets and deeper networks are effective, we should also give some thought to simple architectural add-ons that extract more data from the information already present in each pixel.

Contributions

David Li contributed to the writing of the introduction section of this paper, while Andy Yan handled the abstract. The background information was researched by David Li. The implementation of the traditional architecture was handled by Andy Yan, who also explained it in the design section. David Li coded the Laplacian layer addition to the architecture, and added the explanation to the design section. The results were handled concurrently by Andy Yan and David Li, who analyzed the results for the traditional and Laplacian sections respectively. The conclusion was written by Andy Yan. Finally, the video slides were handled by Andy Yan and David Li, while David Li also did the video presentation.

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